

Chapter 1

Introduction

1.1 Introduction

The fourth-third millennia BC cal (henceforth all dates BC cal will be indicated as BC) in the Near East saw the emergence of the worlds first large urban centres and states, well documented in Mesopotamia and Egypt ^{1.1}. Areas adjacent to Mesopotamia and Egypt, such as the Levant and Anatolia, witnessed significant contemporary changes but debate has developed over whether this represents urbanism, a situation of secondary state formation, or alternative social worlds (e.g. Falconer 1995, Fall et al. 1998, Finkelstein and Gophna 1993, Finkelstein 1995, Flannery 1972, Richard 1987). Whilst this thesis is not directly concerned with providing answers to such questions, these developments, whatever their precise nature, raise important questions about how such fourth-third millennia communities engaged with and impacted upon their landscapes and the extent to which new sorts of landscape transformations occurred which in part reflected the distinctive identities of new sorts of communities.

In particular, the Early Bronze Age (EBA) of the southern Levant, c. 3600 - 1900 BC, has long been regarded as an important transitional period of increasing social complexity (Joffe 1991, 3 and Philip 2001, 163) that witnessed the emergence of larger settlements, fortifications, and large public buildings. These developments have been conventionally interpreted as indicators of the first urban communities in the Southern Levant. A number of authors have argued that these settled communities represent small-scale versions of the fourth and third millennium city-states in southern Mesopotamia and northern Syria (e.g. Ben-Tor 1992a). More recently, it has been suggested that the communities of the Southern Levant are no more than ‘corporate villages’ (discussed in Chapter 2), however, their development still suggests significant degrees of modification of the landscape by these communities and that for the first time communities ‘inscribe’ themselves on the landscape (Chesson 2000, Philip 2001). These propositions have neither been examined closely in the light of archaeological

^{1.1}Cowgill (2004, 526-528) provides an excellent discussion of many of the issues with the use of the terms ‘urban’ and ‘urbanism’

data concerning land use, nor have effective methodologies for the examination of such issues been employed. This thesis aims to redress this imbalance.

The EBA in the Levant witnessed widespread intensified production of what has become known as the classic triad of the Mediterranean crop economy (namely viticulture, olive and cereal production) (Fall et al. 2002, 451-454). The mass production of olive oil and wine in this period is supported by pottery finds (Stager 1985, 176-177), and by palynological studies, which have indicated a marked increase of olive production coinciding with decreasing natural forest cover (Baruch 1986). The nature of the interrelationships between settlement growth and distribution, and the appearance of these combined agricultural practices is an important issue (Esse 1991, Finkelstein and Gophna 1993) that will be explored in this thesis. This thesis aims to examine the significance of the growth of a network of large dense communities in terms of their exploitation of, engagement with, and impact upon their landscapes and to develop a new methodology for investigating these issues by creating models of likely sustaining areas in the landscape around sites and of the agricultural strategies and land use patterns employed by those communities. It is shown in this thesis that examining sustaining catchment areas and land use patterns required for the production of a settlement's crops and animals allows for a more nuanced understanding of settlement growth, site distribution, interactions between communities and potential impacts on the environment.

Changes in subsistence strategies and settlement patterns are largely determined by people's ability to adapt and by their ability to manipulate environmental constraints, both of which can be linked to social, political and economic organisation (Esse 1991, 1). The research presented in this thesis furthers our understanding of the significance of developments from the 5th to the 3rd millennia by examining these changes and what may have prompted them. Using subsistence scenarios based on the archaeological record a computer program has been developed which calculates minimum and maximum catchment areas and maps relevant catchments most proximate to sites studied. Over one hundred sites in the region are modelled using these methods and the results incorporated into a path network/territory analysis. The systematic analysis of the likely sustaining areas of many broadly contemporary sites, together with their potential travel route network and approximate footprint sizes, results in maps which can indicate the degree of landscape exploitation, level of overall impact, possible areas where communities had to interact closely in relationship to access of areas of the landscape, and thus ultimately contribute to an analysis of the way in which people and communities were interacting in the southern Levant during the Chalcolithic and EBA periods. It thus goes some way toward answering questions about how these settlements were organised, both socially and economically, their status as urban centres, and the way in which they engaged with the landscape. In turn, this will provide

new insights into whether EBA settlements were truly organised in a fundamentally different ('urban') way to those of the Chalcolithic.

1.2 Research Questions

As I have indicated the Early Bronze Age saw the appearance of a significant number and density of large settlements, to a degree not witnessed earlier anywhere in the world except Mesopotamia. In the light of the issues and approach I have outlined in Section 1.1, I have developed a number of specific questions, answers to which should provide insights into these phenomena.

It is proposed that agricultural intensification (Philip 2001) must have been associated with the development of these communities; is it possible to find evidence supporting this proposition? Did they outgrow the capacity of the immediately adjacent local landscape to support themselves?

Given the new larger scale of EBA communities, what sort of impact on the landscape can we understand such communities to have had? Are these likely to have been some of the first communities to have had a significant negative impact on their physical environment? Can we find evidence that, as ? put it, such communities 'inscribed' themselves on their landscape in new ways?

Much debate exists as to whether we should see such communities as 'urban'. If we understand urban communities as (at least partly) requiring access to resources to sustain them beyond those afforded by their immediate hinterland, then insight into the above question of whether these Early Bronze Age communities could be sustained by their immediately adjacent local landscape will surely be a valuable contribution to this debate. Can an empirically rigorous argument be made that, contrary to the main trend in scholarly literature on the Early Bronze Age over the past 15 years, landpressures under conditions of population growth in the EBA favoured the development of regional site dependencies, exchange networks and perhaps also settlement hierarchies?

In addressing the above issues, other questions arise. How did such relatively large communities sustain themselves within the environments in which they operated? What sorts of land use can we interpret from the available data on subsistence practices combined with environmental evidence and our understanding of their landscapes? If we can ascertain the types of land use, what scale of activity relating to the associated land use can we understand to have operated in these circumstances?

The use of site catchment analysis helps address several of these questions, but previous 'site catchment' studies have generally only focussed on individual settlements; however, settlements and communities exist in networks. In particular, one of the notable features of the Early Bronze settlement record is the obvious existence of networks

of communities, especially in the larger communities (Philip 2001, 207). What can a combined study of land use and routes connecting a network of settlements allow us to understand about relationships between settlements and the nature of such networks?

The above questions generate important methodological and practical issues. In particular, an effective and efficient way to model multiple scenarios of land use from archaeological and landscape data to compensate for the imprecision inherent in the archaeological record has to be developed. Additionally, an effective way to model land use in reference to a network of settlements will be required.

1.3 Case Study

It is rare to find a site in the Southern Levant that provides a good chronological record of continuous occupation spanning from the Chalcolithic to beyond the EB I. Tell esh Shuna North (henceforth simply Shuna) is one of these rare sites, with a sequence of archaeology from c. 3900-3000 BC cal. Regardless of whether or not one believes that the EB I represents a critical transition from village-level society leading to an ‘urbanised’ society of the EBII (discussed in Chapter 2), this period remains important, because, as previously stated, even if one believes ‘urbanism’ is not the product, it is nevertheless still a period of important change. In addition to its unbroken chronology for the periods of interest to this thesis, Shuna was a large settlement (~ 20 hectares by the Late Early Bronze Age I) that was located at an important junction of major north-south and east-west trade routes (Philip and Baird 1993, 15) so an understanding of this settlement’s subsistence strategies will enhance the understanding of its role in the regional network analysis carried out in Chapter 6.

Furthermore, excavations of the site conducted by Baird and Philip in the early 1990’s provide detailed faunal and palaeobotanical records, crucial for the reconstruction of diet and land use, which is, as the reader will see, a critical component of my methodology. These factors make Shuna an ideal Chalcolithic - EBA settlement with which to explore the development of large ‘urban’ centres, to assess the scale of their impact on the environment, and the degree to which EBA settlements of the southern Levant were interacting both intra-regionally and inter-regionally.

1.4 Thesis Structure

The structure of this thesis is organised so as to guide the reader from the questions posed by research on the Early Bronze Age, through considerations of the methodology needed to answer such questions, development of a methodology original to this research, application of the methodology in case studies of the EB I, to reflections on the effectiveness of the methodology and provisional answers to questions posed in this chapter in the conclusions of Chapters 5, 6 and 7. More specifically this chapter is

followed by Chapter 2, which provides a brief overview of the archaeology of the Early Bronze Age defining in greater detail the questions outlined above and which provided the starting point for this research. A detailed discussion of the geographical setting and climate of the southern Levant is also presented, which provides a firm foundation upon which the subsequent computer modelling and landscape analysis work are based.

Every attempt is made in this thesis to guide the reader through the complicated computer-based methodology in a logical, step-wise, and sequential fashion. The research in this thesis draws heavily upon the use of Geographic Information Systems (GIS) and Chapter 3 begins by providing the reader with a broad overview of its use and application in archaeology, as well as definitions for much of the specific GIS terminology that is necessarily used throughout this thesis. A thorough explanation of the methodology applied to the EBA of the southern Levant is the focus of Chapter 4.

Chapter 4 introduces Landuse Analyst; the software application which was specifically written in order to tackle the research questions addressed by this thesis. Landuse Analyst automates much of the methodology and allows for a fast, efficient and systematic solution to an otherwise very complicated and time-intensive process. No extant commercially available or open source software allows the intricate levels of modelling that was required for this research. Consequently, a considerable component of this project was the development of open source software that provides an interactive, fully functioning, complex (but easy to use) application for modelling land use. Importantly, as will become evident, this software will be shown to be a valuable tool which can be employed beyond the specific archaeological research presented in this thesis. The Appendix of this thesis includes the complete source code for Landuse Analyst[©]. This has been included to illustrate the complexity of this software and to ensure that even if the online code hosting site ceases to exist, this open source software will still be available to anyone wishing to use it. The complete set of files and resources necessary to compile this software is currently available for download online at <http://code.google.com/p/landuseanalyst>.

Having presented how the landscape modelling process works, the efficacy of Landuse Analyst is then demonstrated by applying it to the site of Tel esh Shuna North (hereafter Shuna) in Chapter 5. Thirty six different landuse scenarios ('set' combinations of various input factors used by Landuse Analyst) are presented for each of the three chronological periods examined; the Chalcolithic, EEBI and LEBI. The results are presented in groups which show the effects that various combinations of input factors have on the resultant landscape use catchment areas. Following the presentation of the catchment areas, four of these scenarios (which are shown to consistently produce maximum and minimum extents) are used to create catchments for an additional 125 EB I sites in the region in Chapter 6. These sites are then analysed in terms of the ways

in which they may have been connected in a network of potential pathways. Finally, the results of the landscape analysis derived using Landuse analyst are overlaid and compared with results obtained using more traditional methods (e.g. XTENT modelling) and the effectiveness of the methodology then discussed.

This thesis concludes with a discussion of the future potential for both the general landscape use methodology employed in this thesis, and specifically for Landuse Analyst, in Chapter 7, along with a summary and conclusions of this research.